MOCHAVI SECURITY ASSESSMENT

DUSK BEP20 Audit Report

Formal review of the verified deployment, exposed control surface, and observable supply-management properties.

PREPARED BY
Mochavi for Dusk

ASSESSMENT DATE
April 22, 2026

NETWORK
BNB Smart Chain mainnet

COMPILER
v0.5.16+commit.9c3226ce

ADDRESS
0xb2bd0749dbe21f623d9baba856d3b0f0e1bfec9c

Scope

Item	Value
Contract	`BEP20Token`
Network	BNB Smart Chain mainnet
Address	`0xb2bd0749dbe21f623d9baba856d3b0f0e1bfec9c`
Verified source	https://bscscan.com/token/0xb2bd0749dbe21f623d9baba856d3b0f0e1bfec9c#code
Reviewed artifact	`DuskBEP20Token.sol` verified-source replica
Compiler	`v0.5.16+commit.9c3226ce`
Optimizer	Enabled (`200` runs)

Methodology and Severity Classification

This assessment consisted of manual review of the verified contract source, examination of the privileged token surface, and live RPC validation of the present owner and total-supply

state. The review focused on issuance controls, ownership model, transfer and approval behavior, and the extent to which bridge or migration backing is enforced by the token contract itself.

Severity ratings in this report are assigned according to the following rubric:

- **Critical**: direct and irrecoverable loss of user funds, unrestricted theft, or catastrophic supply compromise requiring no privileged assumptions beyond ordinary network conditions.
- **High**: unbacked minting, privileged-role seizure, or other defects that can materially compromise token integrity or user balances under reachable conditions.
- **Medium**: meaningful accounting or control-plane weaknesses, or architectural trust dependencies that can produce security-relevant impact but do not amount to immediate catastrophic failure.
- **Low**: narrow-scope issues, standard integration hazards, or hardening concerns with limited direct impact.
- **Informational**: descriptive observations, deployment-context notes, or non-defective implementation characteristics recorded for completeness.

Assumptions and Scope Limitations

The official Dusk documentation describes two complementary flows:

- native DUSK on Dusk mainnet may be bridged to BEP20 DUSK on BNB Smart Chain by locking native DUSK on Dusk and minting an equivalent BEP20 amount on BSC; and
- ERC20/BEP20 DUSK may be migrated to native DUSK by locking the EVM-side tokens in a smart contract and issuing native DUSK on Dusk mainnet.

Per project-provided context on April 22, 2026, the BEP20 bridge and the mainnet migration process are now operated as a two-way pathway between Dusk mainnet and BNB Smart Chain.

That context materially affects the interpretation of observed supply growth: post-deployment minting is not, in itself, evidence of unauthorized inflation. However, the migration contracts, bridge operators, and the corresponding Dusk-mainnet accounting logic were not part of this review scope. The present assessment therefore evaluates only those security properties that are enforced by the BEP20 contract itself.

Executive Summary

The BNB Smart Chain DUSK token differs materially from the Ethereum deployment. At the token-contract layer it is not fixed supply: it inherits `Ownable`, exposes an owner-restricted `mint(uint256)` function, and permits ownership transfer or renunciation. Under the project context supplied for this review, the mint mechanism is intended to support a

bidirectional bridge and migration system between Dusk mainnet and BNB Smart Chain rather than discretionary supply expansion.

As of April 22, 2026, a live RPC query yielded the following values:

- `owner() = 0xe6D0c4d8CA2C12b357A5b5470c53E86123Ab16da`
- `totalSupply() = 184077200894984685000000009`
- Human-readable current supply: `184,077,200.894984685000000009 DUSK`
- Constructor supply: `50,000,000 DUSK`
- Net post-deployment minted supply: `134,077,200.894984685000000009 DUSK`

Within this architecture, the central technical observation is that the token contract does not independently validate burns, bridge attestations, or issuance ceilings before minting. Supply parity therefore depends on external governance, reconciliation, and custody procedures. The code is otherwise compact and does not contain fee logic, blacklist logic, proxy hooks, or reentrancy-prone external calls.

Security Properties Summary

Check	Status	Notes
External minting	Contextual	`mint(uint256)` is owner callable and appears intended for migration / bridge issuance.
Privileged owner controls	Caution	`Ownable` governs minting and ownership transfer.
Ownership state	Active	A live owner is still set as of April 22, 2026 to support ongoing issuance flows.
Upgradeability / proxy	Pass	No proxy, delegatecall, or upgrade hooks detected.
Blacklist / pause / fee logic	Pass	No transfer taxes, pause flags, or blacklist logic detected.
Reentrancy surface	Pass	No external calls are made during token state transitions.
Allowance safety	Caution	Direct `approve()` preserves the conventional ERC20/BEP20 allowance race condition.

Findings Summary

Severity	Count
Critical	0
High	0
Medium	1
Low	2
Informational	0

Detailed Findings

M-01: Bridge-backed issuance is not enforced by this token contract

Severity: MEDIUM
Location: `DuskBEP20Token.sol:297`

The contract exposes `mint(uint256 amount)` as an external owner-only function. That function calls `_mint()` and increases both `totalSupply` and the owner's balance. No hard cap, timelock, multisignature requirement, or on-chain issuance predicate is present within the token contract.

Per project-provided context, that mint path is intended to mirror the bridge and migration accounting used when assets move between Dusk mainnet and BNB Smart Chain. The review therefore does not treat post-deployment mint activity as inherently suspicious. The narrower technical issue is that this token contract does not itself enforce any proof of lock, burn, bridge attestation, or issuance cap. Whether each mint is properly backed is therefore determined entirely outside this contract.

The mint mechanism is active. The constructor sets the initial supply to `50,000,000 DUSK` at `DuskBEP20Token.sol:233`, whereas the live supply observed on April 22, 2026 is `184,077,200.894984685000000009 DUSK`.

Impact: If the migration operator were compromised or were to deviate from the intended process, this contract could mint unbacked BEP20 DUSK without any internal on-chain constraint.
Recommendation: If migration remains active, place mint authority behind a contract-enforced bridge controller or a multisignature regime with reconciliation procedures, delay mechanisms, and an auditable issuance policy.

L-01: Active mint authority is currently held by an externally owned account

Severity: LOW
Location: `DuskBEP20Token.sol:183`

The contract uses a standard `Ownable` model and, as of April 22, 2026, `owner()` resolves to `0xe6D0c4d8CA2C12b357A5b5470c53E86123Ab16da`. A direct code query on that address returned `0x`, indicating that the owner is an externally owned account rather than a contract-controlled wallet.

Given the migration context, an active owner is expected. The residual concern is therefore operational hardening: mint authority depends on a single private-key boundary rather than multisignature or contract-enforced governance.

Impact: Single-key compromise or insider misuse can directly trigger the mint path.
Recommendation: Move ownership to a multisig or governance-controlled contract if minting authority must remain active.

L-02: Standard allowance race remains on direct `approve()`

Severity: LOW
Location: `DuskBEP20Token.sol:276`

Like many ERC20/BEP20 implementations, this contract permits direct non-zero-to-non-zero allowance changes through `approve()`. A spender may exploit transaction ordering to consume the previous allowance before the new allowance takes effect.

The contract also includes `increaseAllowance()` and `decreaseAllowance()` at `DuskBEP20Token.sol:287` and `DuskBEP20Token.sol:292`, which provide a safer pattern for incremental updates.

Impact: Allowance front-running remains possible for integrations using raw `approve()`.
Recommendation: Prefer allowance adjustment helpers or the two-step zero-then-set pattern.

Notes

- The BNB-chain contract remains administratively centralized even though it is not technically upgradeable.
- No public burn function exists in the token itself; burn / migration semantics therefore appear to live outside this contract.
- The Solidity compiler version is consistent with the historical deployment period and is not treated in this report as a standalone adverse finding.

Conclusion

Within the stated migration design, the BNB-chain DUSK contract does not present a critical code-level exploit surface. The principal security condition is instead architectural: supply correctness depends on off-contract migration and bridge controls because the token contract itself does not validate backing before minting. Accordingly, the security posture of this deployment depends not only on the token contract, but also on the integrity of the operational processes and privileged custody framework governing issuance.

Sources

- BscScan verified source: https://bscscan.com/token/0xb2bd0749dbe21f623d9baba856d3b0f0e1bfec9c#code
- BscScan contract page: https://bscscan.com/address/0xb2bd0749dbe21f623d9baba856d3b0f0e1bfec9c
- Dusk docs, Native DUSK to BEP20 Bridge: https://docs.dusk.network/learn/guides/bep20-bridge/
- Dusk docs, ERC20/BEP20 Migration: https://docs.dusk.network/learn/guides/mainnet-migration/